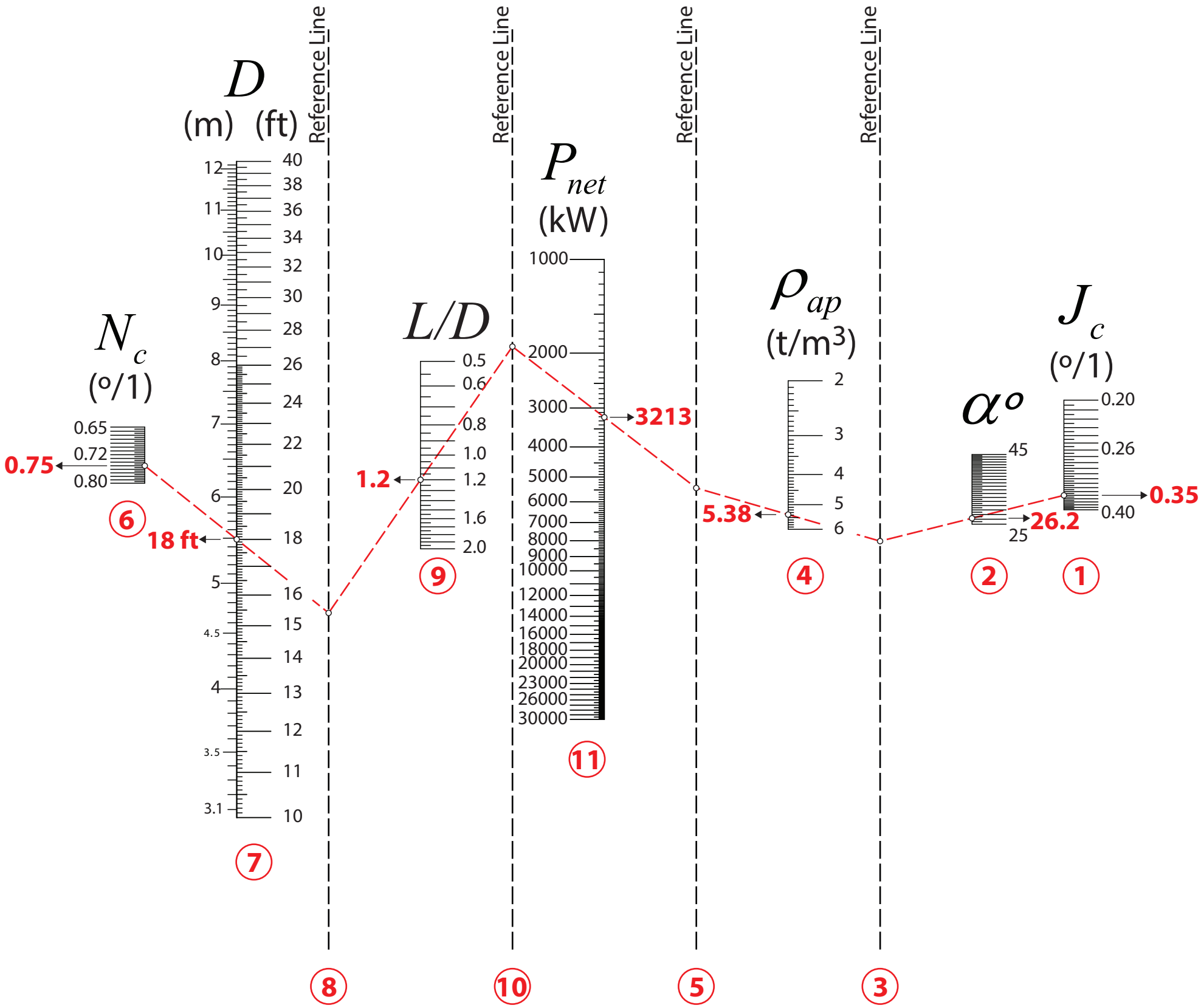
# Modified Hogg & Fuerstenau Mill Power Model




$$P_{net} = 0.267 \cdot D^{3.5} \cdot {}^{L}\!/_{D} \cdot N_c \cdot \rho_{ap} \cdot \left(J_c - J_c^2\right) \cdot \sin\alpha$$

$P_{net}$ = Net mill power draw, kW

$D$ = Effective mill diameter, ft

$L$ = Effective internal mill length, ft

$N_c$ = Rotational mill speed, expressed as a fraction (º/1) of its critical full centrifugation speed

$\rho_{ap}$ = Apparent mill charge density, t/m$^3$

$J_c$ = Apparent mill filling, º/1

$\alpha$ = Lift angle of the charge kidney's center of gravity

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